



Hidden riches

A beach at low tide can look completely empty, aside from the shorebirds picking their way over the sand. But below the surface, it is often teeming with life. Many of the animals are burrowing worms and shellfish that process the sand for edible particles. Others emerge from the sand at high tide to gather plankton from the water, risking attack by predatory fish.



▲ HEART URCHIN

When buried, a heart urchin uses its long tube feet to open up breathing and feeding channels in the sand.

HUNGRY WORMS

At low tide, many sandy beaches are dotted with the coiled casts of burrowing lugworms. These marine worms live in U-shaped burrows that allow them to draw water in at one end. They feed by swallowing sand, digesting any edible material, and ejecting the rest on the beach surface. These casts are swept away every time the tide covers them, so each one indicates the feeding activity of just a few hours.



SPINY BURROWERS

Among the animals that spend their lives hidden in beach sand are heart urchins, sometimes known as sea potatoes. These relatives of typical sea urchins have short, mobile spines that they use for digging, and long, flexible tube feet like those of starfish. They live in burrows in the wet sand, gathering and feeding on the tiny fragments of dead marine life.

WOW!

Eleven square feet (one square meter) of a sandy tidal beach can contain up to 20,000 buried sand mason worms.